

Revealing endogenous conditions for Peto's paradox via an ordinary differential equation model.

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Cancer, a disease intimately linked to cellular mutations, is commonly believed to exhibit a positive association with the cell count and lifespan of a species. Despite this assumption, the observed uniformity in cancer rates across species, referred to as the Peto's paradox, presents a conundrum. Recognizing that tumour progression is not solely dependent on cancer cells but involves intricate interactions among various cell types, this study employed a Lotka-Volterra (LV) ordinary differential equation model to analyze the evolution of cancerous cells and the cancer incidence in an immune environment. As a result, this study uncovered the sufficient conditions underlying the absence of correlation in Peto's paradox and provide insights into the reasons for the equitable distribution of cancer incidence across diverse species by applying nondimensionalization and drawing an analogy between the characteristic time interval for the variation of cell populations in the ODE model and that of cell cycles of a species.

Beyond Peto's paradox: expanding the study of cancer resistance across species.

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Peto's paradox, which highlights the lower-than-expected cancer rates in larger and/or longer-lived species, is a cornerstone of discussions at the intersection of ecology, evolution, and cancer research. It prompts investigations into how species with traits that theoretically increase cancer risk manage to exhibit cancer resistance, with the ultimate goal of uncovering novel therapies for humans. Building on these foundational insights, we propose expanding the research focus to species that, despite possessing traits beyond size and longevity that theoretically increase their cancer risk, exhibit unexpected cancer resistance. Testing Peto's paradox without interference from transient dynamics also requires considering species that are at an equilibrium between cancer risks and defenses, which is increasingly challenging due to anthropogenic activities. Additionally, we argue that transmissible cancers could significantly help in understanding how the metastatic process might be naturally suppressed. This research perspective is timely and aims to support the continued and in-depth identification of anti-cancer adaptations retained throughout evolution in the animal kingdom.

Peto's paradox and human cancers.

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Peto's paradox is the lack of the expected trend in cancer incidence as a function of body size and lifespan across species. The leading hypothesis to explain this pattern is natural selection for differential cancer prevention in larger, longer lived species. We evaluate whether a similar effect exists within species, specifically humans. We begin by reanalysing a recently published dataset to separate the effects of stem cell number and replication rate, and show that each has an independent effect on cancer risk. When considering the lifetime number of stem cell divisions in an extended dataset, and removing cases associated with other diseases or carcinogens, we find that lifetime cancer risk per tissue saturates at approximately 0.3-1.3% for the types considered. We further demonstrate that grouping by anatomical

site explains most of the remaining variation. Our results indicate that cancer risk depends not only on the number of stem cell divisions but varies enormously (approx. 10 000 times) depending on anatomical site. We conclude that variation in risk of human cancer types is analogous to the paradoxical lack of variation in cancer incidence among animal species and may likewise be understood as a result of evolution by natural selection.

Cell size and cancer: a new solution to Peto's paradox?

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Cancer, one of the leading health concerns for humans, is by no means a human-unique malady. Accumulating evidence shows that cancer kills domestic and wild animals at a similar rate to humans and can even pose a conservation threat to certain species. Assuming that each physiologically active and proliferating cell is at risk of malignant transformation, any evolutionary increase in the number of cells (and thus body mass) will lead to a higher cancer frequency, all else being equal. However, available data fail to support the prediction that bigger animals are affected by cancer more than smaller ones. The unexpected lack of correlation between body size (and life span) and cancer risk across taxa was dubbed Peto's paradox. In this perspective, several plausible explanations of Peto's paradox are presented, with the emphasis on a largely underappreciated relation of cell size to both metabolism and cell division rates across species, which we believe are key factors underlying the paradox. We conclude that larger organisms have bigger and slowly dividing cells with lower energy turnover, all significantly reducing the risk of cancer initiation. Solving Peto's paradox will enhance our understanding the evolution of cancer and may provide new implications for cancer prevention and treatment.

Peto's paradox and the hallmarks of cancer: constructing an evolutionary framework for understanding the incidence of cancer.

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An evolutionary perspective can help unify disparate observations and make testable predictions. We consider an evolutionary model in relation to two mechanistic frameworks of cancer biology: multistage carcinogenesis and the hallmarks of cancer. The multistage model predicts that cancer risk increases with body size and longevity; however, this is not observed across species (Peto's paradox), but the paradox is resolved by invoking the evolution of additional genetic mechanisms to suppress cancer in large, long-lived species. It is when cancer cells overcome these defence mechanisms that they exhibit the hallmarks of cancer, driving the ongoing evolution of these defences, which in turn is expected to create the differences observed in the genetics of cancer across species and tissues. To illustrate the utility of an evolutionary model we examined some recently published data linking stem-cell divisions and cancer incidence across a range of tissues and show why the original analysis was faulty, and demonstrate that the data are consistent with a multistage model varying from three to seven mutational hits across different tissues. Finally, we demonstrate how an evolutionary model can both define patterns of inherited (familial) cancer and explain the prevalence of cancer in post-reproductive years, including the dominance of epithelial cancers.

Can Peto's paradox be used as the null hypothesis to identify the role of evolution in natural resistance to cancer? A critical review.

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Carcinogenesis affects not only humans but almost all metazoan species. Understanding the rules driving the occurrence of cancers in the wild is currently expected to provide crucial insights into identifying how some species may have evolved efficient cancer resistance mechanisms. Recently the absence of correlation across species between cancer prevalence and body size (coined as Peto's paradox) has attracted a lot of attention. Indeed, the disparity between this null hypothesis, where every cell is assumed to have an identical probability to undergo malignant transformation, and empirical observations is particularly important to understand, due to the fact that it could facilitate the identification of animal species that are more resistant to carcinogenesis than expected. Moreover it would open up ways to identify the selective pressures that may be involved in cancer resistance. However, Peto's paradox relies on several questionable assumptions, complicating the interpretation of the divergence between expected and observed cancer incidences. Here we review and challenge the different hypotheses on which this paradox relies on with the aim of identifying how this null hypothesis could be better estimated in order to provide a standard protocol to study the deviation between theoretical/theoretically predicted and observed cancer incidence. We show that due to the disproportion and restricted nature of available data on animal cancers, applying Peto's hypotheses at species level could result in erroneous conclusions, and actually assume the existence of a paradox. Instead of using species level comparisons, we propose an organ level approach to be a more accurate test of Peto's assumptions. The accuracy of Peto's paradox assumptions are rarely valid and/or quantifiable, suggesting the need to reconsider the use of Peto's paradox as a null hypothesis in identifying the influence of natural selection on cancer resistance mechanisms.

A metabolic perspective of Peto's paradox and cancer.

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The frequency of cancer is postulated to be proportional to the number of cells an animal possesses, as each cell is similarly exposed to mutagens with every cell division. Larger animals result from more cell divisions with more mutagenic exposure, and hence are expected to have higher frequencies of cancer. Yet, as stipulated by Peto's paradox, larger animals do not have the higher rates of cancers seen in smaller animals despite the significant differences in cell numbers and a longer lifetime that would expose larger animals to more mutagens. The rates of cancer appear to be inversely proportional to animal body size, which scales inversely with specific metabolic rates of mammals. Studies over the past 20 years have linked oncogenes and tumour suppressors to alterations in cancer metabolism, and conversely, mutations in metabolic genes have been documented to trigger tumorigenesis. The by-products and intermediates of metabolism, such as reactive oxygen species, oxoglutarate, citrate and acetate, all have the potential to mutate and alter the genome or epigenome. On the basis of these general observations, it is proposed that metabolic rates correlate with mutagenic rates, which are higher in small animals and give the mechanistic basis for Peto's paradox. The observations discussed in this overview collectively indicate that specific metabolic rate varies inversely with body size, which seems to support the hypothesis that metabolism drives tumorigenesis and accounts for Peto's paradox.

Solutions to Peto's paradox revealed by mathematical modelling and cross-species cancer gene analysis.

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Whales have 1000-fold more cells than humans and mice have 1000-fold fewer; however, cancer risk across species does not increase with the number of somatic cells and the lifespan of the organism. This observation is known as Peto's paradox. How much would evolution have to change the parameters of somatic evolution in order to equalize the cancer risk between species that differ by orders of magnitude in size? Analysis of previously published models of colorectal cancer suggests that a two- to three-fold decrease in the mutation rate or stem cell division rate is enough to reduce a whale's cancer risk to that of a human. Similarly, the addition of one to two required tumour-suppressor gene mutations would also be sufficient. We surveyed mammalian genomes and did not find a positive correlation of tumour-suppressor genes with increasing body mass and longevity. However, we found evidence of the amplification of TP53 in elephants, MAL in horses and FBXO31 in microbats, which might explain Peto's paradox in those species. Exploring parameters that evolution may have fine-tuned in large, long-lived organisms will help guide future experiments to reveal the underlying biology responsible for Peto's paradox and guide cancer prevention in humans.

Natural resistance to cancers: a Darwinian hypothesis to explain Peto's paradox.

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Peto's paradox stipulates that there is no association between body mass (a surrogate of number of cells and longevity) and cancer prevalence in wildlife species. Resolving this paradox is a very promising research direction to understand mechanisms of cancer resistance. As of present, research has been focused on the consequences of these evolutionary pressures rather than of their causes. Here, we argue that evolution through natural selection may have shaped mechanisms of cancer resistance in wildlife species and that this can result in a threshold in body mass above which oncogenic and tumor suppressive mechanisms should be increasingly purified and positively selected, respectively. We conclude that assessing wildlife species in their natural ecosystems, especially through theoretical modeling, is the most promising way to understand how evolutionary processes can favor one or the other pathway. This will provide important insights into mechanisms of cancer resistance.

Peto's Paradox: how has evolution solved the problem of cancer prevention?

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The risk of developing cancer should theoretically increase with both the number of cells and the lifespan of an organism. However, gigantic animals do not get more cancer than humans, suggesting that super-human cancer suppression has evolved numerous times across the tree of life. This is the essence and promise of Peto's Paradox. We discuss what is known about Peto's Paradox and provide hints of what is yet to be discovered.